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(*c*) In a truss strut 3 m long consists of two angles ISA 100 × 100 × 6 m. Find the factored strength if the angles are connected on both sides of 12 mm gusset by one bolt only. 7

(*d*) Solve the above question if welding is used to make a rigid joint. 7

5. (*a*) Distinguish between laterally restrained and unrestrained beams. 2

(*b*) Describe the design procedure for bending strength of a laterally supported beam. 7

(*c*) Describe shear strength and deflection limits checking procedure of a laterally supported beam. 7

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(*d*) Discuss the design procedure of a laterally unsupported beam with check all criteria. 7

320651(20)

BE (6th Semester)

Examination, April-May, 2016

(New Scheme)

Structural Engineering Design-II

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) Use of IS 800 : 2007 is permitted. Draw suitable sketches in each design question.

(*iii*) The figures in the right-hand margin indicate marks.

1. (*a*) What is Partial Safety Factor? 2

(*b*) What is structural steel? Explain List out the important properties of such steel. 7

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(*c*) Explain how limit state method differs from working stress method of design. 7

(*d*) Determine the plastic moment capacity and plastic modulus of section shown in figure below : 7

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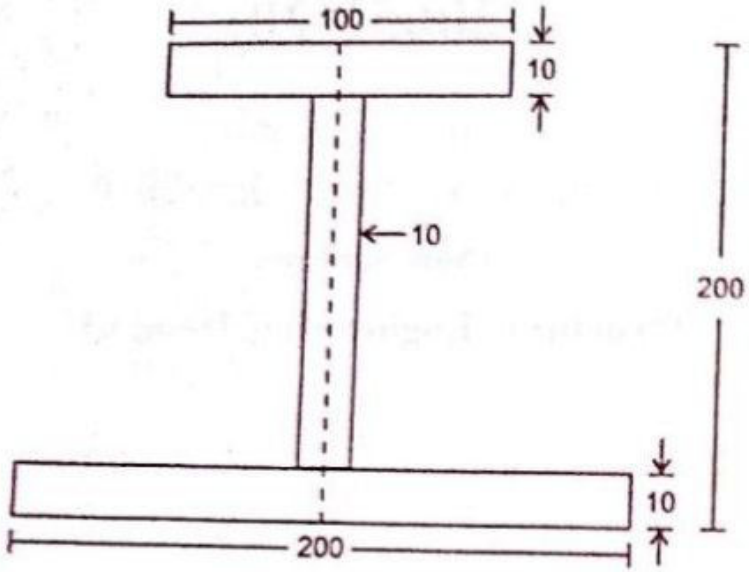

2. (*a*) What is HSFG bolts? 2

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(*b*) Discuss the advantages and disadvantages of bearing bolts and HSFG bolts. 7

(*c*) Neatly sketch the following welded connections : 7

(*i*) Butt weld (groove weld) - single V, double V

(*ii*) Fillet weld

(*iii*) Slot weld, Pug weld

(*d*) Design a suitable fillet weld for a bracket if working load p = 100 kN and eccentricity e = 150 mm. The thickness of the bracket plate is 12 mm and the column used is ISHB 300 @ 618 N/m. 7

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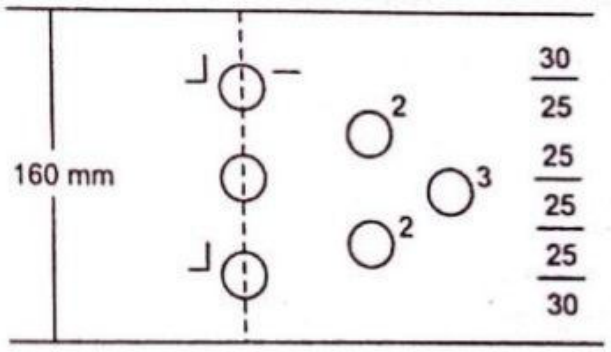
3. (*a*) What is Lug angle? 2

(*b*) Write a short note on block shear failure. 7

(*c*) Write a short note on tension member splices. 7

(*d*) Determine the design tensile strength of 160 × 8 mm plate with the holes for 16 mm bolts as shown in the figure below. Plates are of steel, grade Fe 415. 7



4. (*a*) Differentiate tension member with compression member. 2

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(*b*) Write a descriptive note on lacing and its types with the help of suitable sketches. 7